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UCI on grant-free PUSCH

Abstract

According to certain embodiments, a method for use in a wireless device for transmitting uplink control information (UCI) on grant-free resources comprises reserving a subset of time/frequency resources of a grant-free uplink channel for transmitting UCI, and upon determining the wireless device has UCI to transmit on the grant-free uplink channel, transmitting the UCI to a network node in the reserved subset of time/frequency resources.

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Background/Summary

PRIORITY (1) This nonprovisional application is a continuation, under 35 U.S.C. § 120, of U.S. patent application Ser. No. 16/096,471 filed on Oct. 25, 2018, which is a U.S. National Stage Filing under 35 U.S.C. § 371 of International Patent Application Serial No. PCT/IB2018/057859 filed Oct. 10, 2018 and entitled "UCI ON GRANT-FREE PUSCH" which claims priority to U.S. Provisional Patent Application No. 62/571,112 filed Oct. 11, 2017, all of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD
(1) Embodiments of the present disclosure are directed to wireless communications and, more particularly, to uplink control information (UCI) on a grant-free physical uplink shared channel (PUSCH).
BACKGROUND
(2) Generally, all terms used herein are to be interpreted according to their ordinary meaning in the relevant technical field, unless a different meaning is clearly given and/or is implied from the context in which it is used. All references to a/an/the element, apparatus, component, means, step, etc. are to be interpreted openly as referring to at least one instance of the element, apparatus, component, means, step, etc., unless explicitly stated otherwise. The steps of any methods disclosed herein do not have to be performed in the exact order disclosed, unless a step is explicitly described as following or preceding another step and/or where it is implicit that a step must follow or precede another step. Any feature of any of the embodiments disclosed herein may be applied to any other embodiment, wherever appropriate. Likewise, any advantage of any of the embodiments may apply to any other embodiments, and vice versa. Other objectives, features and advantages of the enclosed embodiments will be apparent from the following description.

(3) In cellular wireless systems, such as Third Generation Partnership Project (3GPP) long term evolution (LTE) and new radio (NR) standards, resources for uplink transmissions are normally scheduled by the network node (eNB, gNB, etc.). The scheduling may be dynamic (i.e., the eNB schedules the uplink transmission per transmission occasion, which may be a transmission time interval (TTI) or multiple TTIs (TTI bundling)). Alternatively, the network node may schedule uplink transmissions using semi persistent scheduling (SPS) where multiple periodic occasions are granted at the same time (i.e., prior to a data transmission). Configuration of SPS includes periodicity of the grant, allocation in time and frequency and modulation and coding scheme (MCS) in subsequent SPS occasions.
(4) One benefit of SPS is latency reduction of uplink data transmissions. Compared to uplink dynamic scheduling, SPS can access the uplink transmission resources more quickly, because SPS removes the sending of scheduling requests by the user equipment (UE) and responding with uplink dynamic grants by the eNB.
(5) To further reduce latency, the periodicity may be reduced to the minimum value (e.g., one TTI in LTE). In prior LTE releases, if the data buffer is empty, then the UE sends padding on the allocated SPS resources. With such a low periodicity, the UE is likely to have empty data, and sending padding at every TTI introduces un-necessary interferences. One solution is skipping uplink data transmissions when the data buffer is empty. The configured resources, however, are still reserved for the UE, which may lead to inefficient resource utilization.
(6) NR includes allocating periodic uplink transmission resources in SPS, which may be referred to as uplink transmission without grant. NR includes additional features to support low-latency and high-reliability requirements.
(7) NR includes two types of uplink transmission without grant (referred to as type 1 and type 2). In type 1 uplink data transmission without grant, resource configuration is only based on RRC (re)configuration without any L1 signaling. Type 2 uplink data transmission without grant is similar to LTE SPS, which is based on both RRC configuration and L1 signaling to activate/deactivate the uplink resources. Because NR specifications are in-progress, the phrases uplink transmission without grant or grant-free, as used herein, both refer to either of the type 1 or type 2 schemes as described above.
(8) Uplink control information (UCI) is carried either by physical uplink control channel (PUCCH) or physical uplink shared channel (PUSCH). The UCI contains one or several uplink control information (i.e., downlink acknowledgement (ACK/NACK), channel quality indicator (CQI), and/or scheduling request (SR)).
(9) UCI is transmitted on PUSCH if the UE transmit user data in the uplink. In this case, PUCCH is not transmitted. When there is no user data to be transmitted, UCI is carried by PUCCH.

SUMMARY

(10) Based on the description above, there currently exist certain challenges for transmission of uplink control information (UCI) when using grant-free resources. For example, in transmission of UCI on grant-based physical uplink shared channel (PUSCH), the resources for UCI are configured by a gNB on the granted PUSCH. There is no mechanism defined, however, for transmitting UCI on grant-free resources.
(11) Particular embodiments provide solutions to these and/or other challenges. For example, particular embodiments include configuration of a part of a grant-free resource for transmission of UCI. In some embodiments, a user equipment (UE) always reserves the UCI resource in grant free PUSCH. In some embodiments, the UE reserves the UCI resource in grant free PUSCH if the UE has UCI to transmit. In general, particular embodiments include configuring a part of grant-free resource for transmission of UCI, where the resources are either always reserved for UCI transmission or only reserved if the UE has UCI to send on PUSCH.
(12) According to some embodiments, a method in a wireless device for transmitting UCI on grant-free resources comprising reserving a subset of time/frequency resources of a grant-free uplink

channel for transmitting UCI, and upon determining the wireless device has UCI to transmit on the grant-free uplink channel, transmitting the UCI to a network node in the reserved subset of time/frequency resources. Upon determining the wireless device does not have UCI to transmit on the grant-free uplink channel, the method may comprise transmitting user data in the reserved subset of time/frequency resources or may comprise not transmitting user data in the reserved subset of time/frequency resources.

(13) In particular embodiments, the method further comprises receiving configuration information for transmitting UCI. The configuration information comprises at least one of the subset of time/frequency resources to reserve for transmitting UCI, UCI payload size; and UCI type. Receiving the configuration information may comprise at least one of receiving radio resource control (RRC) signaling and receiving layer one signaling.

(14) In particular embodiments, the method further comprises indicating whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by transmitting a first demodulation reference signal (DMRS) to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and transmitting a second DMRS to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI. In some embodiments, the method comprises indicating whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by selecting a first pool of resources to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and selecting a second pool of resources to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI.

(15) According to some embodiments, a wireless device is operable to transmit UCI on grant-free resources. The wireless device comprises processing circuitry operable to reserve a subset of time/frequency resources of a grant-free uplink channel for transmitting UCI, and upon determining the wireless device has UCI to transmit on the grant-free uplink channel, transmit the UCI to a network node in the reserved subset of time/frequency resources.

(16) In particular embodiments, the processing circuitry is further operable to, upon determining the wireless device does not have UCI to transmit on the grant-free uplink channel, transmit user data in the reserved subset of time/frequency resources or not transmit user data in the reserved subset of time/frequency resources.

(17) In particular embodiments, the processing circuitry is further operable to receive configuration information for transmitting UCI. The configuration information comprises at least one of the subset of time/frequency resources to reserve for transmitting UCI, UCI payload size; and UCI type. The processing circuitry is operable to receive the configuration information by receiving at least one of RRC signaling and layer one signaling.

(18) In particular embodiments, the processing circuitry is further operable to indicate whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by transmitting a first DMRS to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and transmitting a second DMRS to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI. In some embodiments, the processing circuitry indicates whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by selecting a first pool of resources to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and selecting a second pool of resources to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI.

(19) According to some embodiments, a method in a network node for receiving UCI on grant-free resources comprises transmitting UCI configuration information to a wireless device. The UCI configuration information includes at least one of an indication of a subset of time/frequency resources of a grant-free uplink channel to reserve for transmitting UCI, UCI payload size and UCI type. The method further comprises receiving the grant-free uplink channel and determining

whether the received grant-free uplink channel includes UCI in a reserved subset of time/frequency resources.

(20) Transmitting the UCI configuration information to the wireless device may comprise at least one of transmitting RRC signaling and transmitting layer one signaling.

(21) In particular embodiments, determining whether the received grant-free uplink channel includes UCI in the reserved subset of time/frequency resources comprises determining whether a received DMRS is of a first type indicating that the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI or is of a second type indicating that the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI. In some embodiments, determining whether the received grant-free uplink channel includes UCI (or user data) in the reserved subset of time/frequency resources comprises blindly decoding the reserved subset of time/frequency resources. In some embodiments, determining whether the received grant-free uplink channel includes UCI in the reserved subset of time/frequency resources comprises determining whether the reserved subset of time/frequency resources belong to a first pool of resources or a second pool of resources.

(22) According to some embodiments, a network node is operable to receive UCI on grant-free resources. The network node comprises processing circuitry operable to transmit UCI configuration information to a wireless device. The UCI configuration information includes at least one of an indication of a subset of time/frequency resources of a grant-free uplink channel to reserve for transmitting UCI, UCI payload size and UCI type. The processing circuitry is further operable to receive the grant-free uplink channel and determine whether the received grant-free uplink channel includes UCI in a reserved subset of time/frequency resources.

(23) The processing circuitry may be operable to transmit the UCI configuration information to the wireless device using at least one of RRC signaling and layer one signaling.

(24) In particular embodiments, the processing circuitry is operable to determine whether the received grant-free uplink channel includes UCI in the reserved subset of time/frequency resources by determining whether a received DMRS is of a first type indicating that the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI or is of a second type indicating that the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI. In some embodiments, the processing circuitry is operable to determine whether the received grant-free uplink channel includes UCI in the reserved subset of time/frequency resources by blindly decoding the reserved subset of time/frequency resources. In some embodiments, the processing circuitry is operable to determine whether the received grant-free uplink channel includes UCI in the reserved subset of time/frequency resources by determining whether the reserved subset of time/frequency resources belong to a first pool of resources or a second pool of resources.

(25) According to some embodiments, a wireless device is operable to transmit UCI on grant-free resources. The wireless device comprises a configuring unit and a transmitting unit. The configuring unit is operable to reserve a subset of time/frequency resources of a grant-free uplink channel for transmitting UCI. The transmitting unit is operable to, upon determining the wireless device has UCI to transmit on the grant-free uplink channel, transmit the UCI to a network node in the reserved subset of time/frequency resources.

(26) According to some embodiments, a network node is operable to receive UCI on grant-free resources. The network node comprises a transmitting unit and a receiving unit. The transmitting unit is operable to transmit UCI configuration information to a wireless device. The UCI configuration information includes at least one of an indication of a subset of time/frequency resources of a grant-free uplink channel to reserve for transmitting UCI, UCI payload size and UCI type. The receiving unit is operable to receive the grant-free uplink channel and determine whether the received grant-free uplink channel includes UCI in a reserved subset of time/frequency resources.

(27) Also disclosed is a computer program product comprising a non-transitory computer readable medium storing computer readable program code, the computer readable program code operable, when executed by processing circuitry to perform any of the methods performed by the wireless device described above.
(28) Another computer program product comprises a non-transitory computer readable medium storing computer readable program code, the computer readable program code operable, when executed by processing circuitry to perform any of the methods performed by the network node described above.
(29) Certain embodiments may provide one or more of the following technical advantages. Particular embodiments facilitate UCI transmission on PUSCH when no granted PUSCH is available.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a more complete understanding of the disclosed embodiments and their features and advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a block diagram illustrating reserved resources for uplink control information (UCI) on grant-free physical uplink shared channel (PUSCH);
(3) FIG. **2** is a block diagram illustrating an example wireless network;
(4) FIG. **3** illustrates an example user equipment, according to certain embodiments;
(5) FIG. **4** is a flowchart illustrating an example method in a user equipment, according to certain embodiments;
(6) FIG. **5** is a flowchart illustrating an example method in a network node, according to certain embodiments;
(7) FIG. **6** illustrates an example wireless device, according to certain embodiments;
(8) FIG. **7** illustrates an example network node, according to certain embodiments;
(9) FIG. **8** illustrates an example virtualization environment, according to certain embodiments;
(10) FIG. **9** illustrates an example telecommunication network connected via an intermediate network to a host computer, according to certain embodiments;
(11) FIG. **10** illustrates an example host computer communicating via a base station with a user equipment over a partially wireless connection, according to certain embodiments;
(12) FIG. **11** is a flowchart illustrating a method implemented, according to certain embodiments;
(13) FIG. **12** is a flowchart illustrating a method implemented in a communication system, according to certain embodiments;
(14) FIG. **13** is a flowchart illustrating another method implemented in a communication system, according to certain embodiments; and
(15) FIG. **14** is a flowchart illustrating another method implemented in a communication system, according to certain embodiments.
DETAILED DESCRIPTION
(16) As described above, there currently exist certain challenges with transmission of uplink control information (UCI) when using grant-free resources. For example, in transmission of UCI on grant-based physical uplink shared channel (PUSCH), the resources for UCI are configured by a gNB on the granted PUSCH. There is no mechanism defined, however, for transmitting UCI on grant-free resources.
(17) Particular embodiments provide solutions to these and/or other challenges. For example, particular embodiments include configuration of a part of a grant-free resource for transmission of UCI. In some embodiments, a user equipment (UE) always reserves the UCI resource in grant free

PUSCH. In some embodiments, the UE reserves the UCI resource in grant free PUSCH if the UE has UCI to transmit. In general, particular embodiments include configuring a part of grant-free resource for transmission of UCI, where the resources are either always reserved for UCI transmission or only reserved if the UE has UCI to send on PUSCH.

(18) Some embodiments are described more fully with reference to the accompanying drawings. Other embodiments, however, are contained within the scope of the subject matter disclosed herein, the disclosed subject matter should not be construed as limited to only the embodiments set forth herein; rather, these embodiments are provided by way of example to convey the scope of the subject matter to those skilled in the art.

(19) According to some embodiments, a part of an uplink grant-free resource is reserved for transmission of UCI. An example is illustrated in FIG. **1**.

(20) FIG. **1** is a block diagram illustrating reserved resources for UCI on grant-free PUSCH. A wireless device is capable of performing uplink transmission using uplink time and frequency resources **3**. A subset of time and frequency resources **3** may be reserved as grant-free PUSCH resources **5**. A subset of grant-free PUSCH resources **5** may be reserved as UCI resources **7**.

(21) When the UE has uplink control data to transmit, the UE may use the reserved UCI resources **7** for transmission and may rate match or puncture the grant-free PUSCH **5**. Otherwise, if the UE does not have any uplink control data, the UE can either use the region for grant-free PUSCH or not use the reserved resources.

(22) Particular embodiments include an indication or configuration of some or all of the following information. Particular embodiments include an indication or configuration of time/frequency resources used for UCI. UCI resources may be configured using semi-static RRC configuration. This may be useful, for example, in type 1 transmission without uplink grant. In some embodiments, the UCI resources are indicated by layer one signaling. As an example, these embodiments may be used in type 2 transmission without uplink grant.

(23) Whether the UE reserves resources or not may be indicated to the gNB using particular time/frequency resources for transmission without grant. It may also be indicated using a modified demodulation reference signal (DMRS) sequence. The DMRS sequence may be modified by using a different cyclic shift or applying another orthogonal cover code (OCC), for example. One DMRS sequence indicates no reserved resource and another DMRS sequence indicates that there is a set of reserved resources within the transmitted grant free resource.

(24) In some embodiments, a gNB blindly decodes the UCI part (assuming there is a cyclic redundancy check (CRC)). Depending on whether the gNB successfully decodes the UCI, the gNB determines whether resources are reserved for the UCI (and thus the rate matching/puncturing scheme for PUSCH data). In some embodiments, the gNB may try to blindly decode data including/excluding the reserved resources.

(25) Particular embodiments may indicate the presence/absence of UCI by the resources that the UE selects from a pool of resources. For example, resources selected from pool A may indicate that UCI is present, and resources selected from pool B indicates that UCI is not present.

(26) Particular embodiments include an indication or configuration of UCI payload. For example, the configuration/indication may include UCI payload size, e.g. the number of ACK/NACK bits, etc. The UCI information can be configured by RRC configuration or indicated by layer one signaling.

(27) Particular embodiments include an indication or configuration of UCI type. Information regarding the UCI types, i.e. A/N, CSI, etc. can also be configured/indicated by RRC configuration or layer one indication.

(28) FIG. **2** illustrates an example wireless network, according to certain embodiments. The wireless network may comprise and/or interface with any type of communication, telecommunication, data, cellular, and/or radio network or other similar type of system. In some embodiments, the wireless network may be configured to operate according to specific standards or

other types of predefined rules or procedures. Thus, particular embodiments of the wireless network may implement communication standards, such as Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), and/or other suitable 2G, 3G, 4G, or 5G standards; wireless local area network (WLAN) standards, such as the IEEE 802.11 standards; and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave and/or ZigBee standards.

(29) Network **106** may comprise one or more backhaul networks, core networks, IP networks, public switched telephone networks (PSTNs), packet data networks, optical networks, wide-area networks (WANs), local area networks (LANs), wireless local area networks (WLANs), wired networks, wireless networks, metropolitan area networks, and other networks to enable communication between devices.

(30) Network node **160** and WD **110** comprise various components described in more detail below. These components work together to provide network node and/or wireless device functionality, such as providing wireless connections in a wireless network. In different embodiments, the wireless network may comprise any number of wired or wireless networks, network nodes, base stations, controllers, wireless devices, relay stations, and/or any other components or systems that may facilitate or participate in the communication of data and/or signals whether via wired or wireless connections.

(31) As used herein, network node refers to equipment capable, configured, arranged and/or operable to communicate directly or indirectly with a wireless device and/or with other network nodes or equipment in the wireless network to enable and/or provide wireless access to the wireless device and/or to perform other functions (e.g., administration) in the wireless network. Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs) and NR NodeBs (gNBs)). Base stations may be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and may then also be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station may be a relay node or a relay donor node controlling a relay. A network node may also include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station may also be referred to as nodes in a distributed antenna system (DAS). Yet further examples of network nodes include multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations (BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), core network nodes (e.g., MSCs, MMEs), O&M nodes, OSS nodes, SON nodes, positioning nodes (e.g., E-SMLCs), and/or MDTs. As another example, a network node may be a virtual network node as described in more detail below. More generally, however, network nodes may represent any suitable device (or group of devices) capable, configured, arranged, and/or operable to enable and/or provide a wireless device with access to the wireless network or to provide some service to a wireless device that has accessed the wireless network.

(32) In FIG. **2**, network node **160** includes processing circuitry **170**, device readable medium **180**, interface **190**, auxiliary equipment **184**, power source **186**, power circuitry **187**, and antenna **162**. Although network node **160** illustrated in the example wireless network of FIG. **1** may represent a device that includes the illustrated combination of hardware components, other embodiments may comprise network nodes with different combinations of components. It is to be understood that a network node comprises any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods disclosed herein. Moreover, while the components of network node **160** are depicted as single boxes located within a larger box, or nested within

multiple boxes, in practice, a network node may comprise multiple different physical components that make up a single illustrated component (e.g., device readable medium **180** may comprise multiple separate hard drives as well as multiple RAM modules).

(33) Similarly, network node **160** may be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which may each have their own respective components. In certain scenarios in which network node **160** comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components may be shared among several network nodes. For example, a single RNC may control multiple NodeB's. In such a scenario, each unique NodeB and RNC pair, may in some instances be considered a single separate network node. In some embodiments, network node **160** may be configured to support multiple radio access technologies (RATs). In such embodiments, some components may be duplicated (e.g., separate device readable medium **180** for the different RATs) and some components may be reused (e.g., the same antenna **162** may be shared by the RATs). Network node **160** may also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **160**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, or Bluetooth wireless technologies. These wireless technologies may be integrated into the same or different chip or set of chips and other components within network node **160**.

(34) Processing circuitry **170** is configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being provided by a network node. The operations performed by processing circuitry **170** may include processing information obtained by processing circuitry **170** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

(35) Processing circuitry **170** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide, either alone or in conjunction with other network node **160** components, such as device readable medium **180**, network node **160** functionality. For example, processing circuitry **170** may execute instructions stored in device readable medium **180** or in memory within processing circuitry **170**. Such functionality may include providing any of the various wireless features, functions, or benefits discussed herein. In some embodiments, processing circuitry **170** may include a system on a chip (SOC).

(36) In some embodiments, processing circuitry **170** may include one or more of radio frequency (RF) transceiver circuitry **172** and baseband processing circuitry **174**. In some embodiments, radio frequency (RF) transceiver circuitry **172** and baseband processing circuitry **174** may be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry **172** and baseband processing circuitry **174** may be on the same chip or set of chips, boards, or units

(37) In certain embodiments, some or all of the functionality described herein as being provided by a network node, base station, eNB or other such network device may be performed by processing circuitry **170** executing instructions stored on device readable medium **180** or memory within processing circuitry **170**. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **170** without executing instructions stored on a separate or discrete device readable medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **170** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **170** alone or to other components of network

node **160**, but are enjoyed by network node **160** as a whole, and/or by end users and the wireless network generally.

(38) Device readable medium **180** may comprise any form of volatile or non-volatile computer readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer-executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **170**. Device readable medium **180** may store any suitable instructions, data or information, including a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **170** and, utilized by network node **160**. Device readable medium **180** may be used to store any calculations made by processing circuitry **170** and/or any data received via interface **190**. In some embodiments, processing circuitry **170** and device readable medium **180** may be considered to be integrated.

(39) Interface **190** is used in the wired or wireless communication of signaling and/or data between network node **160**, network **106**, and/or WDs **110**. As illustrated, interface **190** comprises port(s)/terminal(s) **194** to send and receive data, for example to and from network **106** over a wired connection. Interface **190** also includes radio front end circuitry **192** that may be coupled to, or in certain embodiments a part of, antenna **162**. Radio front end circuitry **192** comprises filters **198** and amplifiers **196**. Radio front end circuitry **192** may be connected to antenna **162** and processing circuitry **170**. Radio front end circuitry may be configured to condition signals communicated between antenna **162** and processing circuitry **170**. Radio front end circuitry **192** may receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **192** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **198** and/or amplifiers **196**. The radio signal may then be transmitted via antenna **162**. Similarly, when receiving data, antenna **162** may collect radio signals which are then converted into digital data by radio front end circuitry **192**. The digital data may be passed to processing circuitry **170**. In other embodiments, the interface may comprise different components and/or different combinations of components.

(40) In certain alternative embodiments, network node **160** may not include separate radio front end circuitry **192**, instead, processing circuitry **170** may comprise radio front end circuitry and may be connected to antenna **162** without separate radio front end circuitry **192**. Similarly, in some embodiments, all or some of RF transceiver circuitry **172** may be considered a part of interface **190**. In still other embodiments, interface **190** may include one or more ports or terminals **194**, radio front end circuitry **192**, and RF transceiver circuitry **172**, as part of a radio unit (not shown), and interface **190** may communicate with baseband processing circuitry **174**, which is part of a digital unit (not shown).

(41) Antenna **162** may include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. Antenna **162** may be coupled to radio front end circuitry **190** and may be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In some embodiments, antenna **162** may comprise one or more omni-directional, sector or panel antennas operable to transmit/receive radio signals between, for example, 2 GHz and 66 GHZ. An omni-directional antenna may be used to transmit/receive radio signals in any direction, a sector antenna may be used to transmit/receive radio signals from devices within a particular area, and a panel antenna may be a line of sight antenna used to transmit/receive radio signals in a relatively straight line. In some instances, the use of more than one antenna may be referred to as MIMO. In certain embodiments, antenna **162** may be separate from network node **160** and may be connectable to network node **160** through an interface or port.

(42) Antenna **162**, interface **190**, and/or processing circuitry **170** may be configured to perform any

receiving operations and/or certain obtaining operations described herein as being performed by a network node. Any information, data and/or signals may be received from a wireless device, another network node and/or any other network equipment. Similarly, antenna **162**, interface **190**, and/or processing circuitry **170** may be configured to perform any transmitting operations described herein as being performed by a network node. Any information, data and/or signals may be transmitted to a wireless device, another network node and/or any other network equipment.

(43) Power circuitry **187** may comprise, or be coupled to, power management circuitry and is configured to supply the components of network node **160** with power for performing the functionality described herein. Power circuitry **187** may receive power from power source **186**. Power source **186** and/or power circuitry **187** may be configured to provide power to the various components of network node **160** in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). Power source **186** may either be included in, or external to, power circuitry **187** and/or network node **160**. For example, network node **160** may be connectable to an external power source (e.g., an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry **187**. As a further example, power source **186** may comprise a source of power in the form of a battery or battery pack which is connected to, or integrated in, power circuitry **187**. The battery may provide backup power should the external power source fail. Other types of power sources, such as photovoltaic devices, may also be used.

(44) Alternative embodiments of network node **160** may include additional components beyond those shown in FIG. **2** that may be responsible for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, network node **160** may include user interface equipment to allow input of information into network node **160** and to allow output of information from network node **160**. This may allow a user to perform diagnostic, maintenance, repair, and other administrative functions for network node **160**.

(45) As used herein, wireless device (WD) refers to a device capable, configured, arranged and/or operable to communicate wirelessly with network nodes and/or other wireless devices. Unless otherwise noted, the term WD may be used interchangeably herein with user equipment (UE). Communicating wirelessly may involve transmitting and/or receiving wireless signals using electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information through air. In some embodiments, a WD may be configured to transmit and/or receive information without direct human interaction. For instance, a WD may be designed to transmit information to a network on a predetermined schedule, when triggered by an internal or external event, or in response to requests from the network. Examples of a WD include, but are not limited to, a smart phone, a mobile phone, a cell phone, a voice over IP (VOIP) phone, a wireless local loop phone, a desktop computer, a personal digital assistant (PDA), a wireless cameras, a gaming console or device, a music storage device, a playback appliance, a wearable terminal device, a wireless endpoint, a mobile station, a tablet, a laptop, a laptop-embedded equipment (LEE), a laptop-mounted equipment (LME), a smart device, a wireless customer-premise equipment (CPE). a vehicle-mounted wireless terminal device, etc. A WD may support device-to-device (D2D) communication, for example by implementing a 3GPP standard for sidelink communication, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X) and may in this case be referred to as a D2D communication device. As yet another specific example, in an Internet of Things (IoT) scenario, a WD may represent a machine or other device that performs monitoring and/or measurements and transmits the results of such monitoring and/or measurements to another WD and/or a network node. The WD may in this case be a machine-to-machine (M2M) device, which may in a 3GPP context be referred to as an MTC device. As one example, the WD may be a UE implementing the 3GPP narrow band internet of things (NB-IoT) standard. Examples of such machines or devices are sensors, metering devices such as power

meters, industrial machinery, or home or personal appliances (e.g. refrigerators, televisions, etc.) personal wearables (e.g., watches, fitness trackers, etc.). In other scenarios, a WD may represent a vehicle or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation. A WD as described above may represent the endpoint of a wireless connection, in which case the device may be referred to as a wireless terminal. Furthermore, a WD as described above may be mobile, in which case it may also be referred to as a mobile device or a mobile terminal.

(46) As illustrated, wireless device **110** includes antenna **111**, interface **114**, processing circuitry **120**, device readable medium **130**, user interface equipment **132**, auxiliary equipment **134**, power source **136** and power circuitry **137**. WD **110** may include multiple sets of one or more of the illustrated components for different wireless technologies supported by WD **110**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, WiMAX, or Bluetooth wireless technologies, just to mention a few. These wireless technologies may be integrated into the same or different chips or set of chips as other components within WD **110**.

(47) Antenna **111** may include one or more antennas or antenna arrays, configured to send and/or receive wireless signals, and is connected to interface **114**. In certain alternative embodiments, antenna **111** may be separate from WD **110** and be connectable to WD **110** through an interface or port. Antenna **111**, interface **114**, and/or processing circuitry **120** may be configured to perform any receiving or transmitting operations described herein as being performed by a WD. Any information, data and/or signals may be received from a network node and/or another WD. In some embodiments, radio front end circuitry and/or antenna **111** may be considered an interface.

(48) As illustrated, interface **114** comprises radio front end circuitry **112** and antenna **111**. Radio front end circuitry **112** comprise one or more filters **118** and amplifiers **116**. Radio front end circuitry **114** is connected to antenna **111** and processing circuitry **120** and is configured to condition signals communicated between antenna **111** and processing circuitry **120**. Radio front end circuitry **112** may be coupled to or a part of antenna **111**. In some embodiments, WD **110** may not include separate radio front end circuitry **112**; rather, processing circuitry **120** may comprise radio front end circuitry and may be connected to antenna **111**. Similarly, in some embodiments, some or all of RF transceiver circuitry **122** may be considered a part of interface **114**. Radio front end circuitry **112** may receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **112** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **118** and/or amplifiers **116**. The radio signal may then be transmitted via antenna **111**. Similarly, when receiving data, antenna **111** may collect radio signals which are then converted into digital data by radio front end circuitry **112**. The digital data may be passed to processing circuitry **120**. In other embodiments, the interface may comprise different components and/or different combinations of components.

(49) Processing circuitry **120** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software, and/or encoded logic operable to provide, cither alone or in conjunction with other WD **110** components, such as device readable medium **130**, WD **110** functionality. Such functionality may include providing any of the various wireless features or benefits discussed herein. For example, processing circuitry **120** may execute instructions stored in device readable medium **130** or in memory within processing circuitry **120** to provide the functionality disclosed herein.

(50) As illustrated, processing circuitry **120** includes one or more of RF transceiver circuitry **122**, baseband processing circuitry **124**, and application processing circuitry **126**. In other embodiments, the processing circuitry may comprise different components and/or different combinations of components. In certain embodiments processing circuitry **120** of WD **110** may comprise a SOC. In

some embodiments, RF transceiver circuitry **122**, baseband processing circuitry **124**, and application processing circuitry **126** may be on separate chips or sets of chips. In alternative embodiments, part or all of baseband processing circuitry **124** and application processing circuitry **126** may be combined into one chip or set of chips, and RF transceiver circuitry **122** may be on a separate chip or set of chips. In still alternative embodiments, part or all of RF transceiver circuitry **122** and baseband processing circuitry **124** may be on the same chip or set of chips, and application processing circuitry **126** may be on a separate chip or set of chips. In yet other alternative embodiments, part or all of RF transceiver circuitry **122**, baseband processing circuitry **124**, and application processing circuitry **126** may be combined in the same chip or set of chips. In some embodiments, RF transceiver circuitry **122** may be a part of interface **114**. RF transceiver circuitry **122** may condition RF signals for processing circuitry **120**.

(51) In certain embodiments, some or all of the functionality described herein as being performed by a WD may be provided by processing circuitry **120** executing instructions stored on device readable medium **130**, which in certain embodiments may be a computer-readable storage medium. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **120** without executing instructions stored on a separate or discrete device readable storage medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **120** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **120** alone or to other components of WD **110**, but are enjoyed by WD **110**, and/or by end users and the wireless network generally.

(52) Processing circuitry **120** may be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being performed by a WD. These operations, as performed by processing circuitry **120**, may include processing information obtained by processing circuitry **120** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored by WD **110**, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

(53) Device readable medium **130** may be operable to store a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **120**. Device readable medium **130** may include computer memory (e.g., Random Access Memory (RAM) or Read Only Memory (ROM)), mass storage media (e.g., a hard disk), removable storage media (e.g., a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **120**. In some embodiments, processing circuitry **120** and device readable medium **130** may be integrated.

(54) User interface equipment **132** may provide components that allow for a human user to interact with WD **110**. Such interaction may be of many forms, such as visual, audial, tactile, etc. User interface equipment **132** may be operable to produce output to the user and to allow the user to provide input to WD **110**. The type of interaction may vary depending on the type of user interface equipment **132** installed in WD **110**. For example, if WD **110** is a smart phone, the interaction may be via a touch screen; if WD **110** is a smart meter, the interaction may be through a screen that provides usage (e.g., the number of gallons used) or a speaker that provides an audible alert (e.g., if smoke is detected). User interface equipment **132** may include input interfaces, devices and circuits, and output interfaces, devices and circuits. User interface equipment **132** is configured to allow input of information into WD **110** and is connected to processing circuitry **120** to allow processing circuitry **120** to process the input information. User interface equipment **132** may include, for example, a microphone, a proximity or other sensor, keys/buttons, a touch display, one or more cameras, a USB port, or other input circuitry. User interface equipment **132** is also

configured to allow output of information from WD **110**, and to allow processing circuitry **120** to output information from WD **110**. User interface equipment **132** may include, for example, a speaker, a display, vibrating circuitry, a USB port, a headphone interface, or other output circuitry. Using one or more input and output interfaces, devices, and circuits, of user interface equipment **132**, WD **110** may communicate with end users and/or the wireless network and allow them to benefit from the functionality described herein.

(55) Auxiliary equipment **134** is operable to provide more specific functionality which may not be generally performed by WDs. This may comprise specialized sensors for doing measurements for various purposes, interfaces for additional types of communication such as wired communications etc. The inclusion and type of components of auxiliary equipment **134** may vary depending on the embodiment and/or scenario.

(56) Power source **136** may, in some embodiments, be in the form of a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic devices or power cells, may also be used. WD **110** may further comprise power circuitry **137** for delivering power from power source **136** to the various parts of WD **110** which need power from power source **136** to carry out any functionality described or indicated herein. Power circuitry **137** may in certain embodiments comprise power management circuitry. Power circuitry **137** may additionally or alternatively be operable to receive power from an external power source; in which case WD **110** may be connectable to the external power source (such as an electricity outlet) via input circuitry or an interface such as an electrical power cable. Power circuitry **137** may also in certain embodiments be operable to deliver power from an external power source to power source **136**. This may be, for example, for the charging of power source **136**. Power circuitry **137** may perform any formatting, converting, or other modification to the power from power source **136** to make the power suitable for the respective components of WD **110** to which power is supplied.

(57) Although the subject matter described herein may be implemented in any appropriate type of system using any suitable components, the embodiments disclosed herein are described in relation to a wireless network, such as the example wireless network illustrated in FIG. **2**. For simplicity, the wireless network of FIG. **2** only depicts network **106**, network nodes **160** and **160***b*, and WDs **110**, **110***b*, and **110***c*. In practice, a wireless network may further include any additional elements suitable to support communication between wireless devices or between a wireless device and another communication device, such as a landline telephone, a service provider, or any other network node or end device. Of the illustrated components, network node **160** and wireless device (WD) **110** are depicted with additional detail. The wireless network may provide communication and other types of services to one or more wireless devices to facilitate the wireless devices' access to and/or use of the services provided by, or via, the wireless network.

(58) FIG. **3** illustrates an example user equipment, according to certain embodiments. As used herein, a user equipment or UE may not necessarily have a user in the sense of a human user who owns and/or operates the relevant device. Instead, a UE may represent a device that is intended for sale to, or operation by, a human user but which may not, or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller). Alternatively, a UE may represent a device that is not intended for sale to, or operation by, an end user but which may be associated with or operated for the benefit of a user (e.g., a smart power meter). UE **200** may be any UE identified by the 3.sup.rd Generation Partnership Project (3GPP), including a NB-IoT UE, a machine type communication (MTC) UE, and/or an enhanced MTC (eMTC) UE. UE **200**, as illustrated in FIG. **3**, is one example of a WD configured for communication in accordance with one or more communication standards promulgated by the 3.sup.rd Generation Partnership Project (3GPP), such as 3GPP's GSM, UMTS, LTE, and/or 5G standards. As mentioned previously, the term WD and UE may be used interchangeable. Accordingly, although FIG. **2** is a UE, the components discussed herein are equally applicable to a WD, and vice-versa.

(59) In FIG. **3**, UE **200** includes processing circuitry **201** that is operatively coupled to input/output interface **205**, radio frequency (RF) interface **209**, network connection interface **211**, memory **215** including random access memory (RAM) **217**, read-only memory (ROM) **219**, and storage medium **221** or the like, communication subsystem **231**, power source **233**, and/or any other component, or any combination thereof. Storage medium **221** includes operating system **223**, application program **225**, and data **227**. In other embodiments, storage medium **221** may include other similar types of information. Certain UEs may utilize all of the components shown in FIG. **3**, or only a subset of the components. The level of integration between the components may vary from one UE to another UE. Further, certain UEs may contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.
(60) In FIG. **3**, processing circuitry **201** may be configured to process computer instructions and data. Processing circuitry **201** may be configured to implement any sequential state machine operative to execute machine instructions stored as machine-readable computer programs in the memory, such as one or more hardware-implemented state machines (e.g., in discrete logic, FPGA, ASIC, etc.); programmable logic together with appropriate firmware; one or more stored program, general-purpose processors, such as a microprocessor or Digital Signal Processor (DSP), together with appropriate software; or any combination of the above. For example, the processing circuitry **201** may include two central processing units (CPUs). Data may be information in a form suitable for use by a computer.
(61) In the depicted embodiment, input/output interface **205** may be configured to provide a communication interface to an input device, output device, or input and output device. UE **200** may be configured to use an output device via input/output interface **205**. An output device may use the same type of interface port as an input device. For example, a USB port may be used to provide input to and output from UE **200**. The output device may be a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. UE **200** may be configured to use an input device via input/output interface **205** to allow a user to capture information into UE **200**. The input device may include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display may include a capacitive or resistive touch sensor to sense input from a user. A sensor may be, for instance, an accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, another like sensor, or any combination thereof. For example, the input device may be an accelerometer, a magnetometer, a digital camera, a microphone, and an optical sensor.
(62) In FIG. **3**, RF interface **209** may be configured to provide a communication interface to RF components such as a transmitter, a receiver, and an antenna. Network connection interface **211** may be configured to provide a communication interface to network **243*a***. Network **243*a*** may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **243*a*** may comprise a Wi-Fi network. Network connection interface **211** may be configured to include a receiver and a transmitter interface used to communicate with one or more other devices over a communication network according to one or more communication protocols, such as Ethernet, TCP/IP, SONET, ATM, or the like. Network connection interface **211** may implement receiver and transmitter functionality appropriate to the communication network links (e.g., optical, electrical, and the like). The transmitter and receiver functions may share circuit components, software or firmware, or alternatively may be implemented separately.
(63) RAM **217** may be configured to interface via bus **202** to processing circuitry **201** to provide storage or caching of data or computer instructions during the execution of software programs such as the operating system, application programs, and device drivers. ROM **219** may be configured to

provide computer instructions or data to processing circuitry **201**. For example, ROM **219** may be configured to store invariant low-level system code or data for basic system functions such as basic input and output (I/O), startup, or reception of keystrokes from a keyboard that are stored in a non-volatile memory. Storage medium **221** may be configured to include memory such as RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, or flash drives. In one example, storage medium **221** may be configured to include operating system **223**, application program **225** such as a web browser application, a widget or gadget engine or another application, and data file **227**. Storage medium **221** may store, for use by UE **200**, any of a variety of various operating systems or combinations of operating systems.

(64) Storage medium **221** may be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), floppy disk drive, flash memory, USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HD-DVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as a subscriber identity module or a removable user identity (SIM/RUIM) module, other memory, or any combination thereof. Storage medium **221** may allow UE **200** to access computer-executable instructions, application programs or the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a communication system may be tangibly embodied in storage medium **221**, which may comprise a device readable medium.

(65) In FIG. **3**, processing circuitry **201** may be configured to communicate with network **243***b* using communication subsystem **231**. Network **243***a* and network **243***b* may be the same network or networks or different network or networks. Communication subsystem **231** may be configured to include one or more transceivers used to communicate with network **243***b*. For example, communication subsystem **231** may be configured to include one or more transceivers used to communicate with one or more remote transceivers of another device capable of wireless communication such as another WD, UE, or base station of a radio access network (RAN) according to one or more communication protocols, such as IEEE 802.2, CDMA, WCDMA, GSM, LTE, UTRAN, WiMax, or the like. Each transceiver may include transmitter **233** and/or receiver **235** to implement transmitter or receiver functionality, respectively, appropriate to the RAN links (e.g., frequency allocations and the like). Further, transmitter **233** and receiver **235** of each transceiver may share circuit components, software or firmware, or alternatively may be implemented separately.

(66) In the illustrated embodiment, the communication functions of communication subsystem **231** may include data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the global positioning system (GPS) to determine a location, another like communication function, or any combination thereof. For example, communication subsystem **231** may include cellular communication, Wi-Fi communication, Bluetooth communication, and GPS communication. Network **243***b* may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **243***b* may be a cellular network, a Wi-Fi network, and/or a near-field network. Power source **213** may be configured to provide alternating current (AC) or direct current (DC) power to components of UE **200**.

(67) The features, benefits and/or functions described herein may be implemented in one of the components of UE **200** or partitioned across multiple components of UE **200**. Further, the features,

benefits, and/or functions described herein may be implemented in any combination of hardware, software or firmware. In one example, communication subsystem **231** may be configured to include any of the components described herein. Further, processing circuitry **201** may be configured to communicate with any of such components over bus **202**. In another example, any of such components may be represented by program instructions stored in memory that when executed by processing circuitry **201** perform the corresponding functions described herein. In another example, the functionality of any of such components may be partitioned between processing circuitry **201** and communication subsystem **231**. In another example, the non-computationally intensive functions of any of such components may be implemented in software or firmware and the computationally intensive functions may be implemented in hardware.

(68) FIG. **4** is a flowchart illustrating an example method in a wireless device, according to certain embodiments. In particular embodiments, one or more steps of FIG. **4** may be performed by wireless device **110** described with respect to FIG. **2**.

(69) The method begins at step **4112** where a wireless device (e.g., wireless device **110**) receives UCI configuration information for transmitting UCI. For example, wireless device **110** may receive UCI configuration information from network node **160**. The configuration information comprises at least one of the subset of time/frequency resources to reserve for transmitting UCI, UCI payload size; and UCI type. Receiving the configuration information may comprise at least one of receiving RRC signaling and receiving layer one signaling.

(70) At step **4114**, the wireless device reserves a subset of time/frequency resources of a grant-free uplink channel for transmitting UC. For example, wireless device **110** may reserve a subset of time/frequency resources as illustrated in FIG. **1**.

(71) The wireless device determines whether it has UCI to transmit with the next uplink transmission. For example, the wireless device may determine it has HARQ responses to transmit. If the wireless device does have UCI to transmit, then the method continues to step **4116** where the wireless device transmits the UCI to a network node in the reserved subset of time/frequency resources. When the wireless device does not have UCI to transmit, the method continues to step **4118** where the wireless device transmits user data (or no data) in the reserved subset of time/frequency resources.

(72) In particular embodiments, the wireless device may indicate whether the subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by transmitting a first DMRS to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and transmitting a second DMRS to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI. In some embodiments, the method comprises indicating whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by selecting a first pool of resources to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and selecting a second pool of resources to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI.

(73) Modifications, additions, or omissions may be made to method **4100** of FIG. **4**. Additionally, one or more steps in the method of FIG. **4** may be performed in parallel or in any suitable order.

(74) FIG. **5** is a flowchart illustrating an example method in a network node, according to certain embodiments. In particular embodiments, one or more steps of FIG. **5** may be performed by network node **160** described with respect to FIG. **2**.

(75) The method begins at step **5112** where a network node (e.g., wireless device **160**) transmits UCI configuration information to a wireless device. The UCI configuration information includes at least one of an indication of a subset of time/frequency resources of a grant-free uplink channel to reserve for transmitting UCI, UCI payload size, and UCI type. Transmitting the UCI configuration information to the wireless device may comprise at least one of transmitting RRC signaling and transmitting layer one signaling.

(76) At step **5114**, the network node receives the grant-free uplink channel. For example, network node **160** may receive an uplink transmission from wireless device **110**. The uplink transmission may include a grant-free uplink channel as illustrated in FIG. **1**.
(77) At step **5116**, the network node determines whether the received grant-free uplink channel includes UCI in a reserved subset of time/frequency resources. For example, network node **160** may blindly decode a reserved subset of time/frequency resources. Blind decoding may comprise blind decoding for UCI on reserved resources or blind decoding for data on data resources including/excluding reserved resources.
(78) In some embodiments, network node **160** may determine whether a received DMRS is of a first type indicating that the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI or is of a second type indicating that the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI. In some embodiments, network node **160** may determine the received grant-free uplink channel includes UCI in the reserved subset of time/frequency resources comprises determining whether the reserved subset of time/frequency resources belong to a first pool of resources or a second pool of resources.
(79) Modifications, additions, or omissions may be made to method **5100** of FIG. **5**. Additionally, one or more steps in the method of FIG. **5** may be performed in parallel or in any suitable order.
(80) FIG. **6** illustrates an example wireless device, according to certain embodiments. The apparatus may be implemented in a wireless device (e.g., wireless device **110** illustrated in FIG. **1**). Apparatus **1600** is operable to carry out the example method described with reference to FIG. **4** and possibly any other processes or methods disclosed herein. It is also to be understood that the method of FIG. **4** is not necessarily carried out solely by apparatus **1600**. At least some operations of the method can be performed by one or more other entities, including virtual apparatuses.
(81) Apparatus **1600** may comprise processing circuitry, which may include one or more microprocessor or microcontrollers, as well as other digital hardware, which may include digital signal processors (DSPs), special-purpose digital logic, and the like. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as read-only memory (ROM), random-access memory, cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein, in several embodiments. In some implementations, the processing circuitry may be used to cause receiving unit **1602**, determining unit **1604**, storage unit **1606**, and communication unit **1608** and any other suitable units of apparatus **1600** to perform corresponding functions according one or more embodiments of the present disclosure.
(82) As illustrated in FIG. **6**, apparatus **1600** includes receiving unit **1602**, configuring unit **1604**, and transmitting unit **1606**. In certain embodiments, receiving unit **1602** may receive UCI configuration from a network node **160** according to any of the embodiments and examples described herein. Configuring unit **1604** may reserve a subset of time/frequency resources of a grant-free uplink channel for transmitting UCI according to any of the embodiments and examples described herein. In certain embodiments, transmitting unit **1608** may transmit a grant-free uplink channel (with or without UCI) to a network node according to any of the embodiments and examples described herein.
(83) FIG. **7** illustrates an example network node, according to certain embodiments. The apparatus may be implemented in a network node (e.g., network node **160** illustrated in FIG. **1**). Apparatus **1700** is operable to carry out the example method described with reference to FIG. **5** and possibly any other processes or methods disclosed herein. It is also to be understood that the method of FIG. **5** is not necessarily carried out solely by apparatus **1700**. At least some operations of the method can be performed by one or more other entities, including virtual apparatuses.
(84) Apparatus **1700** may comprise processing circuitry, which may include one or more

microprocessor or microcontrollers, as well as other digital hardware, which may include digital signal processors (DSPs), special-purpose digital logic, and the like. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as read-only memory (ROM), random-access memory, cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein, in several embodiments. In some implementations, the processing circuitry may be used to cause receiving unit **1702** and determining unit **1704** and any other suitable units of apparatus **1700** to perform corresponding functions according one or more embodiments of the present disclosure.
(85) As illustrated in FIG. **7**, apparatus **1700** includes configuring unit **1702**, receiving unit **1704**, and transmitting unit **1706**. In certain embodiments, configuring unit **1702** may transmit UCI configuration information to a wireless device according to any of the embodiments and examples described herein. In certain embodiments, receiving unit **1704** may receive an uplink transmission from a wireless device according to any of the embodiments and examples described herein. In certain embodiments, transmitting unit **1704** may, in conjunction with configuring unit **1702**, transmit UCI configuration information to a wireless device according to any of the embodiments and examples described herein.
(86) FIG. **8** is a schematic block diagram illustrating a virtualization environment **300** in which functions implemented by some embodiments may be virtualized. In the present context, virtualizing means creating virtual versions of apparatuses or devices which may include virtualizing hardware platforms, storage devices and networking resources. As used herein, virtualization can be applied to a node (e.g., a virtualized base station or a virtualized radio access node) or to a device (e.g., a UE, a wireless device or any other type of communication device) or components thereof and relates to an implementation in which at least a portion of the functionality is implemented as one or more virtual components (e.g., via one or more applications, components, functions, virtual machines or containers executing on one or more physical processing nodes in one or more networks).
(87) In some embodiments, some or all of the functions described herein may be implemented as virtual components executed by one or more virtual machines implemented in one or more virtual environments **300** hosted by one or more of hardware nodes **330**. Further, in embodiments in which the virtual node is not a radio access node or does not require radio connectivity (e.g., a core network node), then the network node may be entirely virtualized.
(88) The functions may be implemented by one or more applications **320** (which may alternatively be called software instances, virtual appliances, network functions, virtual nodes, virtual network functions, etc.) operative to implement some of the features, functions, and/or benefits of some of the embodiments disclosed herein. Applications **320** are run in virtualization environment **300** which provides hardware **330** comprising processing circuitry **360** and memory **390**. Memory **390** contains instructions **395** executable by processing circuitry **360** whereby application **320** is operative to provide one or more of the features, benefits, and/or functions disclosed herein.
(89) Virtualization environment **300**, comprises general-purpose or special-purpose network hardware devices **330** comprising a set of one or more processors or processing circuitry **360**, which may be commercial off-the-shelf (COTS) processors, dedicated Application Specific Integrated Circuits (ASICs), or any other type of processing circuitry including digital or analog hardware components or special purpose processors. Each hardware device may comprise memory **390-1** which may be non-persistent memory for temporarily storing instructions **395** or software executed by processing circuitry **360**. Each hardware device may comprise one or more network interface controllers (NICs) **370**, also known as network interface cards, which include physical network interface **380**. Each hardware device may also include non-transitory, persistent, machine-readable storage media **390-2** having stored therein software **395** and/or instructions executable by

processing circuitry **360**. Software **395** may include any type of software including software for instantiating one or more virtualization layers **350** (also referred to as hypervisors), software to execute virtual machines **340** as well as software allowing it to execute functions, features and/or benefits described in relation with some embodiments described herein.

(90) Virtual machines **340**, comprise virtual processing, virtual memory, virtual networking or interface and virtual storage, and may be run by a corresponding virtualization layer **350** or hypervisor. Different embodiments of the instance of virtual appliance **320** may be implemented on one or more of virtual machines **340**, and the implementations may be made in different ways.

(91) During operation, processing circuitry **360** executes software **395** to instantiate the hypervisor or virtualization layer **350**, which may sometimes be referred to as a virtual machine monitor (VMM). Virtualization layer **350** may present a virtual operating platform that appears like networking hardware to virtual machine **340**.

(92) As shown in FIG. **8**, hardware **330** may be a standalone network node with generic or specific components. Hardware **330** may comprise antenna **3225** and may implement some functions via virtualization. Alternatively, hardware **330** may be part of a larger cluster of hardware (e.g. such as in a data center or customer premise equipment (CPE)) where many hardware nodes work together and are managed via management and orchestration (MANO) **3100**, which, among others, oversees lifecycle management of applications **320**.

(93) Virtualization of the hardware is in some contexts referred to as network function virtualization (NFV). NFV may be used to consolidate many network equipment types onto industry standard high volume server hardware, physical switches, and physical storage, which can be located in data centers, and customer premise equipment.

(94) In the context of NFV, virtual machine **340** may be a software implementation of a physical machine that runs programs as if they were executing on a physical, non-virtualized machine. Each of virtual machines **340**, and that part of hardware **330** that executes that virtual machine, be it hardware dedicated to that virtual machine and/or hardware shared by that virtual machine with others of the virtual machines **340**, forms a separate virtual network elements (VNE).

(95) Still in the context of NFV, Virtual Network Function (VNF) is responsible for handling specific network functions that run in one or more virtual machines **340** on top of hardware networking infrastructure **330** and corresponds to application **320** in FIG. **8**.

(96) In some embodiments, one or more radio units **3200** that each include one or more transmitters **3220** and one or more receivers **3210** may be coupled to one or more antennas **3225**. Radio units **3200** may communicate directly with hardware nodes **330** via one or more appropriate network interfaces and may be used in combination with the virtual components to provide a virtual node with radio capabilities, such as a radio access node or a base station.

(97) In some embodiments, some signaling can be effected with the use of control system **3230** which may alternatively be used for communication between the hardware nodes **330** and radio units **3200**.

(98) With reference to FIG. **9**, in accordance with an embodiment, a communication system includes telecommunication network **410**, such as a 3GPP-type cellular network, which comprises access network **411**, such as a radio access network, and core network **414**. Access network **411** comprises a plurality of base stations **412***a*, **412***b*, **412***c*, such as NBs, eNBs, gNBs or other types of wireless access points, each defining a corresponding coverage area **413***a*, **413***b*, **413***c*. Each base station **412***a*, **412***b*, **412***c* is connectable to core network **414** over a wired or wireless connection **415**. A first UE **491** located in coverage area **413***c* is configured to wirelessly connect to, or be paged by, the corresponding base station **412***c*. A second UE **492** in coverage area **413***a* is wirelessly connectable to the corresponding base station **412***a*. While a plurality of UEs **491**, **492** are illustrated in this example, the disclosed embodiments are equally applicable to a situation where a sole UE is in the coverage area or where a sole UE is connecting to the corresponding base station **412**.

(99) Telecommunication network **410** is itself connected to host computer **430**, which may be embodied in the hardware and/or software of a standalone server, a cloud-implemented server, a distributed server or as processing resources in a server farm. Host computer **430** may be under the ownership or control of a service provider or may be operated by the service provider or on behalf of the service provider. Connections **421** and **422** between telecommunication network **410** and host computer **430** may extend directly from core network **414** to host computer **430** or may go via an optional intermediate network **420**. Intermediate network **420** may be one of, or a combination of more than one of, a public, private or hosted network; intermediate network **420**, if any, may be a backbone network or the Internet; in particular, intermediate network **420** may comprise two or more sub-networks (not shown).
(100) The communication system of FIG. **9** as a whole enables connectivity between the connected UEs **491**, **492** and host computer **430**. The connectivity may be described as an over-the-top (OTT) connection **450**. Host computer **430** and the connected UEs **491**, **492** are configured to communicate data and/or signaling via OTT connection **450**, using access network **411**, core network **414**, any intermediate network **420** and possible further infrastructure (not shown) as intermediaries. OTT connection **450** may be transparent in the sense that the participating communication devices through which OTT connection **450** passes are unaware of routing of uplink and downlink communications. For example, base station **412** may not or need not be informed about the past routing of an incoming downlink communication with data originating from host computer **430** to be forwarded (e.g., handed over) to a connected UE **491**. Similarly, base station **412** need not be aware of the future routing of an outgoing uplink communication originating from the UE **491** towards the host computer **430**.
(101) FIG. **10** illustrates an example host computer communicating via a base station with a user equipment over a partially wireless connection, according to certain embodiments. Example implementations, in accordance with an embodiment, of the UE, base station and host computer discussed in the preceding paragraphs will now be described with reference to FIG. **10**. In communication system **500**, host computer **510** comprises hardware **515** including communication interface **516** configured to set up and maintain a wired or wireless connection with an interface of a different communication device of communication system **500**. Host computer **510** further comprises processing circuitry **518**, which may have storage and/or processing capabilities. In particular, processing circuitry **518** may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. Host computer **510** further comprises software **511**, which is stored in or accessible by host computer **510** and executable by processing circuitry **518**. Software **511** includes host application **512**. Host application **512** may be operable to provide a service to a remote user, such as UE **530** connecting via OTT connection **550** terminating at UE **530** and host computer **510**. In providing the service to the remote user, host application **512** may provide user data which is transmitted using OTT connection **550**.
(102) Communication system **500** further includes base station **520** provided in a telecommunication system and comprising hardware **525** enabling it to communicate with host computer **510** and with UE **530**. Hardware **525** may include communication interface **526** for setting up and maintaining a wired or wireless connection with an interface of a different communication device of communication system **500**, as well as radio interface **527** for setting up and maintaining at least wireless connection **570** with UE **530** located in a coverage area (not shown in FIG. **10**) served by base station **520**. Communication interface **526** may be configured to facilitate connection **560** to host computer **510**. Connection **560** may be direct, or it may pass through a core network (not shown in FIG. **10**) of the telecommunication system and/or through one or more intermediate networks outside the telecommunication system. In the embodiment shown, hardware **525** of base station **520** further includes processing circuitry **528**, which may comprise one or more programmable processors, application-specific integrated circuits, field

programmable gate arrays or combinations of these (not shown) adapted to execute instructions. Base station **520** further has software **521** stored internally or accessible via an external connection.
(103) Communication system **500** further includes UE **530** already referred to. Its hardware **535** may include radio interface **537** configured to set up and maintain wireless connection **570** with a base station serving a coverage area in which UE **530** is currently located. Hardware **535** of UE **530** further includes processing circuitry **538**, which may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. UE **530** further comprises software **531**, which is stored in or accessible by UE **530** and executable by processing circuitry **538**. Software **531** includes client application **532**. Client application **532** may be operable to provide a service to a human or non-human user via UE **530**, with the support of host computer **510**. In host computer **510**, an executing host application **512** may communicate with the executing client application **532** via OTT connection **550** terminating at UE **530** and host computer **510**. In providing the service to the user, client application **532** may receive request data from host application **512** and provide user data in response to the request data. OTT connection **550** may transfer both the request data and the user data. Client application **532** may interact with the user to generate the user data that it provides.
(104) It is noted that host computer **510**, base station **520** and UE **530** illustrated in FIG. **10** may be similar or identical to host computer **430**, one of base stations **412***a*, **412***b*, **412***c* and one of UEs **491**, **492** of FIG. **9**, respectively. This is to say, the inner workings of these entities may be as shown in FIGS. **4** and **5** and independently, the surrounding network topology may be that of FIG. **9**.
(105) In FIG. **10**, OTT connection **550** has been drawn abstractly to illustrate the communication between host computer **510** and UE **530** via base station **520**, without explicit reference to any intermediary devices and the precise routing of messages via these devices. Network infrastructure may determine the routing, which it may be configured to hide from UE **530** or from the service provider operating host computer **510**, or both. While OTT connection **550** is active, the network infrastructure may further take decisions by which it dynamically changes the routing (e.g., on the basis of load balancing consideration or reconfiguration of the network).
(106) Wireless connection **570** between UE **530** and base station **520** is in accordance with the teachings of the embodiments described throughout this disclosure. One or more of the various embodiments improve the performance of OTT services provided to UE **530** using OTT connection **550**, in which wireless connection **570** forms the last segment. More precisely, the teachings of these embodiments may improve the signaling overhead and reduce latency, which may provide faster internet access for users.
(107) A measurement procedure may be provided for the purpose of monitoring data rate, latency and other factors on which the one or more embodiments improve. There may further be an optional network functionality for reconfiguring OTT connection **550** between host computer **510** and UE **530**, in response to variations in the measurement results. The measurement procedure and/or the network functionality for reconfiguring OTT connection **550** may be implemented in software **511** and hardware **515** of host computer **510** or in software **531** and hardware **535** of UE **530**, or both. In embodiments, sensors (not shown) may be deployed in or in association with communication devices through which OTT connection **550** passes; the sensors may participate in the measurement procedure by supplying values of the monitored quantities exemplified above or supplying values of other physical quantities from which software **511**, **531** may compute or estimate the monitored quantities. The reconfiguring of OTT connection **550** may include message format, retransmission settings, preferred routing etc.; the reconfiguring need not affect base station **520**, and it may be unknown or imperceptible to base station **520**. Such procedures and functionalities may be known and practiced in the art. In certain embodiments, measurements may involve proprietary UE signaling facilitating host computer **510**'s measurements of throughput,

propagation times, latency and the like. The measurements may be implemented in that software **511** and **531** causes messages to be transmitted, in particular empty or 'dummy' messages, using OTT connection **550** while it monitors propagation times, errors etc.

(108) FIG. **11** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **9** and **10**. For simplicity of the present disclosure, only drawing references to FIG. **11** will be included in this section. In step **610**, the host computer provides user data. In substep **611** (which may be optional) of step **610**, the host computer provides the user data by executing a host application. In step **620**, the host computer initiates a transmission carrying the user data to the UE. In step **630** (which may be optional), the base station transmits to the UE the user data which was carried in the transmission that the host computer initiated, in accordance with the teachings of the embodiments described throughout this disclosure. In step **640** (which may also be optional), the UE executes a client application associated with the host application executed by the host computer.

(109) FIG. **12** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **9** and **10**. For simplicity of the present disclosure, only drawing references to FIG. **12** will be included in this section. In step **710** of the method, the host computer provides user data. In an optional substep (not shown) the host computer provides the user data by executing a host application. In step **720**, the host computer initiates a transmission carrying the user data to the UE. The transmission may pass via the base station, in accordance with the teachings of the embodiments described throughout this disclosure. In step **730** (which may be optional), the UE receives the user data carried in the transmission.

(110) FIG. **13** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **9** and **10**. For simplicity of the present disclosure, only drawing references to FIG. **13** will be included in this section. In step **810** (which may be optional), the UE receives input data provided by the host computer. Additionally or alternatively, in step **820**, the UE provides user data. In substep **821** (which may be optional) of step **820**, the UE provides the user data by executing a client application. In substep **811** (which may be optional) of step **810**, the UE executes a client application which provides the user data in reaction to the received input data provided by the host computer. In providing the user data, the executed client application may further consider user input received from the user. Regardless of the specific manner in which the user data was provided, the UE initiates, in substep **830** (which may be optional), transmission of the user data to the host computer. In step **840** of the method, the host computer receives the user data transmitted from the UE, in accordance with the teachings of the embodiments described throughout this disclosure.

(111) FIG. **14** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **9** and **10**. For simplicity of the present disclosure, only drawing references to FIG. **14** will be included in this section. In step **910** (which may be optional), in accordance with the teachings of the embodiments described throughout this disclosure, the base station receives user data from the UE. In step **920** (which may be optional), the base station initiates transmission of the received user data to the host computer. In step **930** (which may be optional), the host computer receives the user data carried in the transmission initiated by the base station.

(112) The term unit may have conventional meaning in the field of electronics, electrical devices and/or electronic devices and may include, for example, electrical and/or electronic circuitry, devices, modules, processors, memories, logic solid state and/or discrete devices, computer

programs or instructions for carrying out respective tasks, procedures, computations, outputs, and/or displaying functions, and so on, as such as those that are described herein.
(113) Generally, all terms used herein are to be interpreted according to their ordinary meaning in the relevant technical field, unless a different meaning is clearly given and/or is implied from the context in which it is used. All references to a/an/the element, apparatus, component, means, step, etc. are to be interpreted openly as referring to at least one instance of the element, apparatus, component, means, step, etc., unless explicitly stated otherwise. The steps of any methods disclosed herein do not have to be performed in the exact order disclosed, unless a step is explicitly described as following or preceding another step and/or where it is implicit that a step must follow or precede another step. Any feature of any of the embodiments disclosed herein may be applied to any other embodiment, wherever appropriate. Likewise, any advantage of any of the embodiments may apply to any other embodiments, and vice versa. Other objectives, features and advantages of the enclosed embodiments will be apparent from the following description.
(114) Modifications, additions, or omissions may be made to the systems and apparatuses disclosed herein without departing from the scope of the invention. The components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses may be performed by more, fewer, or other components. Additionally, operations of the systems and apparatuses may be performed using any suitable logic comprising software, hardware, and/or other logic. As used in this document, “each” refers to each member of a set or each member of a subset of a set.
(115) Modifications, additions, or omissions may be made to the methods disclosed herein without departing from the scope of the invention. The methods may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order.
(116) The foregoing description sets forth numerous specific details. It is understood, however, that embodiments may be practiced without these specific details. In other instances, well-known circuits, structures and techniques have not been shown in detail in order not to obscure the understanding of this description. Those of ordinary skill in the art, with the included descriptions, will be able to implement appropriate functionality without undue experimentation.
(117) References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments, whether or not explicitly described.
(118) Although this disclosure has been described in terms of certain embodiments, alterations and permutations of the embodiments will be apparent to those skilled in the art. Accordingly, the above description of the embodiments does not constrain this disclosure. Other changes, substitutions, and alterations are possible without departing from the spirit and scope of this disclosure, as defined by the claims below.
(119) At least some of the following abbreviations may be used in this disclosure. If there is an inconsistency between abbreviations, preference should be given to how it is used above. If listed multiple times below, the first listing should be preferred over any subsequent listing(s).
(120) TABLE-US-00001 Abbreviation Explanation 1x RTT CDMA2000 1x Radio Transmission Technology 3GPP 3rd Generation Partnership Project 5G 5th Generation ABS Almost Blank Subframe ARQ Automatic Repeat Request AWGN Additive White Gaussian Noise BCCH Broadcast Control Channel BCH Broadcast Channel BLER Block Error Rate CA Carrier Aggregation CC Carrier Component CCCH SDU Common Control Channel SDU CDMA Code Division Multiplexing Access CGI Cell Global Identifier CIR Channel Impulse Response CP

Cyclic Prefix CPICH Common Pilot Channel CPICH Ec/No CPICH Received energy per chip divided by the power density in the band CQI Channel Quality information C-RNTI Cell RNTI CSI Channel State Information DCCH Dedicated Control Channel DCI Downlink Control Information DFTS OFDM Discrete Fourier Transform Spread OFDM DL Downlink DM Demodulation DMRS Demodulation Reference Signal DRX Discontinuous Reception DTX Discontinuous Transmission DTCH Dedicated Traffic Channel DUT Device Under Test E-CID Enhanced Cell-ID (positioning method) E-SMLC Evolved-Serving Mobile Location Centre ECGI Evolved CGI eNB E-UTRAN NodeB ePDCCH enhanced Physical Downlink Control Channel E-SMLC evolved Serving Mobile Location Center E-UTRA Evolved UTRA E-UTRAN Evolved UTRAN FDD Frequency Division Duplex GERAN GSM EDGE Radio Access Network gNB Base station in NR GNSS Global Navigation Satellite System GSM Global System for Mobile communication HARQ Hybrid Automatic Repeat Request HO Handover HSPA High Speed Packet Access HRPD High Rate Packet Data IR-HARQ Incremental Redundancy HARQ L1 Layer 1 LLR Log Likelihood Ratio LOS Line of Sight LPP LTE Positioning Protocol LTE Long-Term Evolution MAC Medium Access Control MBMS Multimedia Broadcast Multicast Services MBSFN Multimedia Broadcast multicast service Single Frequency Network MBSFN ABS MBSFN Almost Blank Subframe MDT Minimization of Drive Tests MIB Master Information Block MME Mobility Management Entity MSC Mobile Switching Center NPDCCH Narrowband Physical Downlink Control Channel NR New Radio OCNG OFDMA Channel Noise Generator OFDM Orthogonal Frequency Division Multiplexing OFDMA Orthogonal Frequency Division Multiple Access OSS Operations Support System OTDOA Observed Time Difference of Arrival O&M Operation and Maintenance PBCH Physical Broadcast Channel P-CCPCH Primary Common Control Physical Channel PCell Primary Cell PCFICH Physical Control Format Indicator Channel PDCCH Physical Downlink Control Channel PDP Profile Delay Profile PDSCH Physical Downlink Shared Channel PDU Protocol Data Unit PGW Packet Gateway PHICH Physical Hybrid-ARQ Indicator Channel PLMN Public Land Mobile Network PMI Precoder Matrix Indicator PRACH Physical Random Access Channel PRS Positioning Reference Signal PSS Primary Synchronization Signal PUCCH Physical Uplink Control Channel PUSCH Physical Uplink Shared Channel RACH Random Access Channel QAM Quadrature Amplitude Modulation RAN Radio Access Network RAT Radio Access Technology RLM Radio Link Management RNC Radio Network Controller RNTI Radio Network Temporary Identifier RRC Radio Resource Control RRM Radio Resource Management RS Reference Signal RSCP Received Signal Code Power RSRP Reference Symbol Received Power OR Reference Signal Received Power RSRQ Reference Signal Received Quality OR Reference Symbol Received Quality RSSI Received Signal Strength Indicator RSTD Reference Signal Time Difference SC Successive Cancellation SCL Successive Cancellation List SCH Synchronization Channel SCell Secondary Cell SDU Service Data Unit SFN System Frame Number SGW Serving Gateway SI System Information SIB System Information Block SNR Signal to Noise Ratio SON Self Optimized Network SPS Semi-Persistent Scheduling SS Synchronization Signal SSB Synchronization Signal Block SSS Secondary Synchronization Signal TDD Time Division Duplex TDOA Time Difference of Arrival TOA Time of Arrival TSS Tertiary Synchronization Signal TTI Transmission Time Interval UCI Uplink Control Information UE User Equipment UL Uplink UMTS Universal Mobile Telecommunication System URLLC Ultra Reliable Low Latency Communication USIM Universal Subscriber Identity Module UTDOA Uplink Time Difference of Arrival UTRA Universal Terrestrial Radio Access UTRAN Universal Terrestrial Radio Access Network V2X Vehicle to everything VoIP Voice over Internet Protocol WCDMA Wide CDMA WLAN Wide Local Area Network

Claims

1. A method in a wireless device for transmitting on grant-free resources, the grant-free resources pertaining to Type 1 and Type 2 Uplink (UL) transmission without grant in New Radio (NR), the method comprising: receiving uplink control information (UCI) configuration information from a network node, the UCI configuration information including at least an indication of a subset of time/frequency resources of a grant-free uplink channel to reserve to transmit UCI; reserving the subset of time/frequency resources of a grant-free uplink channel for transmitting UCI; and when the wireless device does not have UCI to transmit on the grant-free uplink channel, transmitting user data in the reserved subset of time/frequency resources.
2. The method of claim 1, when the wireless device has UCI to transmit on the grant-free uplink channel, transmitting the UCI to a network node in the reserved subset of time/frequency resources.
3. The method of claim 1, wherein receiving the UCI configuration information comprises at least one of receiving radio resource control (RRC) signaling and receiving layer one signaling.
4. The method of claim 1, further comprising indicating whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by transmitting a first demodulation reference signal (DMRS) to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and transmitting a second DMRS to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI.
5. The method of claim 1, further comprising indicating whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by selecting a first pool of resources to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and selecting a second pool of resources to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI.
6. A wireless device operable to transmit on grant-free resources, the grant-free resources pertaining to Type 1 and Type 2 Uplink (UL) transmission without grant in New Radio (NR), the wireless device comprising processing circuitry operable to: receive uplink control information (UCI) configuration information from a network node, the UCI configuration information including at least an indication of a subset of time/frequency resources of a grant-free uplink channel to reserve to transmit UCI; reserve the subset of time/frequency resources of a grant-free uplink channel for transmitting UCI; and when the wireless device does not have UCI to transmit on the grant-free uplink channel, transmit user data in the reserved subset of time/frequency resources.
7. The wireless device of claim 6, the processing circuitry further operable to, when the wireless device has UCI to transmit on the grant-free uplink channel, transmit the UCI to a network node in the reserved subset of time/frequency resources.
8. The wireless device of claim 6, wherein the processing circuitry is operable to receive the UCI configuration information by receiving at least one of radio resource control (RRC) signaling and layer one signaling.
9. The wireless device of claim 6, the processing circuitry further operable to indicate whether a subset of time/frequency resources of a grant-free uplink channel are reserved for transmitting UCI by transmitting a first demodulation reference signal (DMRS) to indicate the grant-free uplink channel includes a subset of time/frequency resources reserved for UCI and transmitting a second DMRS to indicate the grant-free uplink channel does not include a subset of time/frequency resources reserved for UCI.